



Chemical tempering of soda lime silicate glasses by ion exchange process for the improvement of surface and bulk mechanical strength



İpek Erdem^a, Duygu Guldiren^b, Suheyla Aydin^{a,*}

^a Istanbul Technical University, Department of Metallurgical and Materials Engineering, Istanbul 34469, Turkey

^b Şişecam Science and Technology Center, Şişecam Str. No. 2, Çayırova, Kocaeli, Turkey

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ABSTRACT

The aim of this work is improving mechanical properties of soda-lime glass by imparting compression stress on the surface through ion exchange and investigate the effect of process parameters on chemical strengthening. KNO₃ salt bath was used for ionic exchange at varying temperatures and durations. Treated samples were investigated for their optical transmittance, surface potassium ionic concentrations and concentration profiles from surface to bulk using EDS and line scan EDS techniques. Effect of process temperature and time on the amount of mechanical strengthening was determined through the bending strengths (four-point), scratch resistances and indentation behaviors as micro hardness, crack formation load limit and cracking probability. Results were evaluated regarding ionic concentration variations. Transmittance decreased by 2% after the treatment. Surface potassium concentration increased with increasing temperature and was increased to become saturated in a short period. Potassium concentrations decreased from surface to bulk, and diffusion reached to deeper layers with increasing time and temperature. Ionic exchange improved micro hardness by 8–17%, threshold loads for indentation crack initiation 2–10 times, cracking probability down to 50%, strength up to 3.5–4 times and scratch resistance 2–4 times. Crack types changed between radial, median and lateral; depending on the degree of chemical tempering.

1. Introduction

With rapidly developing technology in our century, glass materials particularly started being used not only as housewares but also in many fields such as automotive, architecture, construction, aircraft, optical and aerospace industries, laboratories, energy generation [1–7]. Glass is the mostly preferred or the only choice for its esthetics and transparency, so to meet the needs of the sector and expand the use in different applications, glasses should be reliable materials with high strength [6]. Therefore, the development concerning mechanical properties of glass materials are of great importance, and today the efforts in this regard continue to rise [2,4,6].

Schneider et al. [8] defines glass as a brittle solid, behaving almost perfectly elastic. Mechanical properties, which are strictly related to the amount, depth and distribution of surface flaws, are the most important features of glass that determines its use. These surface micro cracks cause the glass to present poor mechanical behavior i.e. tensile strength varying within the tens of MPa, even though theoretically its structure enables up to thousands of MPa [3,8–12]. Studies on glass mechanical strengthening focuses on the suppression of these micro cracks by

introducing compressive stresses to the glass surface, where stress limits the generation and propagation of cracks [1,7,10,11,13]. Besides other methods for inducing surface compression such as, thermal tempering, ion exchange, ion bombardment and surface crystallization and many more [3,11,14,15] chemical tempering has become the focus of studies in the last decades.

Ion exchange method used for chemical tempering is also applied for many years in production of optical waveguides, glass coloring and gaining antibacterial properties to glass [4]. In recent years, chemical tempering has become a popular choice for developing glass strength and improving the surface properties of glass. There are a variety of products and patents developed by different glass manufacturers in this regard; and still researches continue considering the effect of parameters on process efficiency such as temperature, time, glass composition and thickness, bath composition and concentration, exchanging pair of ions, specimen geometry, quality of glass surface etc. [4,5,11]. Since each parameter has numerous variables and affects the results to a great extent, each study carries its own authenticity. Chemical tempering is realized by the ionic exchange process of the alkali ions in the glass structure with the ions in a molten salt bath it is

* Corresponding author.

E-mail address: saydin@itu.edu.tr (S. Aydin).

in contact with. The larger ions incorporated to the surface structure replacing the smaller ions cause local strains thus increase the strength of glass by forming a compressive stress [1,4,10–14,16]. This method can be applied easily to complex shaped and very thin (down to 0.5 mm) parts of varying sizes, the resulting product is suitable for machining and the process does not require expensive machinery or equipment [1,3,10]. The majority of the products commercialized by chemical tempering method use alkali aluminosilicate glass; developed products generally serve as electronic equipment panels and screens.

In the present study, it was aimed to improve the mechanical properties of soda-lime glass by chemical tempering and investigate the effect of ion exchange process parameters on strengthening. Ion exchange was applied using pure KNO_3 salt bath and soda-lime silicate glass at varying temperatures and durations. Samples were examined for their optical properties, hardness values and surface ionic concentrations using EDS technique. A process duration and a temperature was selected to investigate the effect of temperature and time on mechanical strengthening. Further characterizations as indentation crack formation behaviors, scratch resistances and bending strengths were realized on selected samples. The results were evaluated together with the surface and bulk ionic concentration variations.

Hardness, cracking probability, deformation zone and scratch resistance analysis give information about the mechanical response of only one surface of glass samples, which were but ion exchanged on both sides under presented experimental studies. Regarding the nature of bending test, the mechanical behavior is determined by both bulk and surface properties of the glass samples. Since both sides of the glass sample contain a certain level of compressive stress sustained by ion exchange, the correlations with surface mechanical properties must be handled carefully. The comparative interpretation of the mechanical test results examined from various angles are thought to be essential contributions to literature, as no studies were found including the mentioned characterization steps all at once.

2. Experimental

2.1. Sample preparation

Experimental studies were conducted on Corning® 2947 soda-lime glass slides of $25 \times 25 \times 1 (\pm 0.01)$ mm. As a result of chemical analysis of the glass realized by a Perkin Elmer Analyst 800 atomic absorption spectrometer with an error estimate of $\pm 2\%$, the composition was determined as (wt.%) 70.56 SiO_2 , 14.54 Na_2O , 8.93 CaO , 3.41 MgO , 0.87 K_2O , 0.59 Al_2O_3 and 1.1 other.

Before the ion exchange process, thermal analysis of the unprocessed glass samples was performed using a PerkinElmer™ Diamond TG/DTA, in platinum pans with a specified weight of 25 mg, from room temperature to 800 °C with a heating rate of 10 °C/min under flowing (100 ml/min) argon gas. The obtained glass transition onset temperature (T_g) from the thermal analysis results was 549 °C. Ion exchange process temperatures were determined accordingly.

Glass samples were cleaned in an ultrasonic cleaner using deionized water and ethanol prior to ionic exchange process to maintain the contamination of the surface at a minimum, increase the contact area and improve the efficiency of ionic exchange process. Cleaning step of the samples were repeated after the ion exchange process in order to remove the salt bath residues from the sample surface.

Ion exchange process on glass samples was performed in a ProthermPLF 120/7 electrical furnace. Ionic exchange was conducted using 100% potassium nitrate salt bath (KNO_3 , 99.0% purity, Alfa Aesar Company). Process was realized at temperatures 425 °C, 450 °C, 475 °C and 500 °C determined as being under the glass transition temperature (T_g) and for 0.5, 1, 2, 4, 8, 12, 16, 20 and 24 h. For narrative simplicity, samples were given notations representing the varying experimental

parameters such as “425-1” representing the sample treated at 425 °C for 1 h.

Since the atomic weight of potassium ion is higher than sodium ion, an increase in sample weight was expected after the ion exchange process. A digital balance of 10^{-4} precision was used for the weight measurements of the ultrasonically cleaned samples before and after the ion exchange process.

2.2. Characterization methods

UV–Vis spectroscopy technique was used to investigate the effect of potassium ion exchange on the optical transmittance of the glasses in the visible region between 360 and 700 nm wavelengths, with a spectral bandwidth 0.1 nm. The investigations were realized using PG Instruments T80 + UV–Vis spectrophotometer, with a tungsten lamp and at room temperature.

The effect of ion exchange on micro hardness of glass samples were investigated through Vickers hardness measurements before and after the process. The indentations were realized under 0.49 N with a constant loading rate for 10 s dwell time using Shimadzu HMV-G21 Vickers micro hardness instrument capable of loading between 0.49 (50 g) to 19.61 N (2 kg). An optical microscope connected to a CCD camera and image analysis software aided for the measurements of the indent diagonals, to give the hardness values. An average value of 20 indentations was determined as the hardness of the sample. The tests were conducted at 23 ± 1 °C with 50–60% relative humidity.

Surface and cross section morphology were determined together with the amount of potassium incorporated to the surface by scanning electron microscopy technique by JEOL™ JCM-6000 Benchtop scanning electron microscope (SEM) and energy dispersive X-ray spectroscopy (EDS) technique using JEOL™ JSM 5410 microscope connected to Noran 2100 Freedom energy dispersive X-ray spectrometer. The concentrations of potassium and sodium ions on the glass surface after the ion exchange process were determined in weight percent. The diffusion depth of incorporated potassium ions after the ionic exchange were also determined using the same instrument by conducting a line scan EDS analysis on selected glass samples.

Schimidzu HMV-G21 Vickers micro hardness instrument was also used to examine the variations of indentation crack resistance of the selected glass samples after ion exchange through determining the variations in threshold loads for indentation crack initiation (TLIC), crack sizes and cracking probabilities, with methods inspired from former studies [9,12]. Applied loads on each of the sample were 0.49 N (50 g), 0.98 N (100 g), 1.96 N (200 g), 2.94 N (300 g), 4.90 N (500 g), 9.80 N (1 kg) and 19.61 N (2 kg) for 10 s dwell time. 20 indentations were realized per load and a crack was accepted to be initiated if at least two of the four corners had started to form cracks. Crack size was determined as the average length of the cracks formed and investigations were realized after 5 min to capture the delayed or slow growing cracks. Indent and crack images were taken under $40 \times$ zoom for loads up to 4.90 N, and under $10 \times$ zoom over that load to capture the full cracks. Cracking probabilities were determined in percentage per each load by calculating the number of indentations resulted in cracking out of the 20 indentations and comparisons were realized for identical loading conditions. Crack type and size comparisons were conducted under 19.61 N (2 kg) load where all selected samples had formed full cracks.

Scratch resistance of samples were examined using CSM nano-indentation and nano-scratch tester using a Rockwell diamond indenter with a 200 μm tip radius, resembling the studies of Bandyopadhyay et al. [17] and Petit et al. [18]. Each scratch test had a pre-scan phase carried at a very low load aiding to determine the sample surface topology, creating a baseline for the main scratching scan. Samples were cleaned with acetone and ethanol before the test. Scratch tests were applied at a constant speed of 0.02 mm/s with load raising gradually to 20 N over a length of 8 mm. Measurements were conducted at room

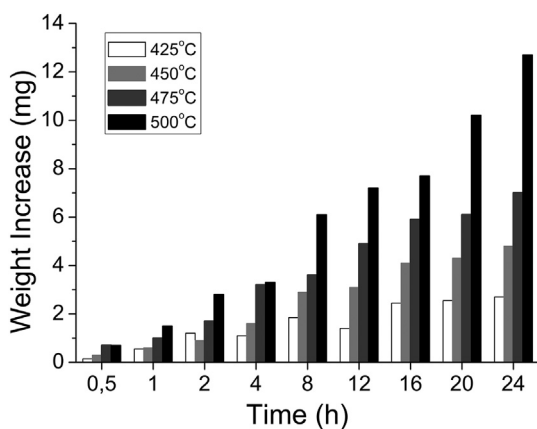


Fig. 1. Weight increase after ion exchange process.

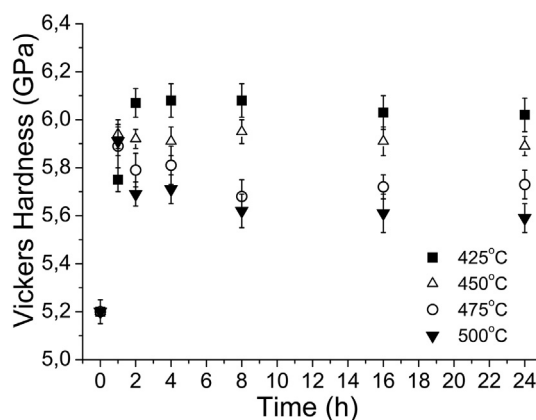


Fig. 3. Micro hardness alterations after the ion exchange process.

temperature and under dry air conditions. The scratch depth variance and acoustic emission spectra was recorded on the sample surface against the applied load.

Strength of ion exchanged glass samples was examined via four point bending test with conditions determined according to ASTM Standard C158-95. For the purpose, samples of $75 \times 25 \times 1$ mm were produced under formerly selected parameters. Schimadzu Universal Testing Instrument was used with an arrangement of inner and outer spans to 10 and 20 mm, respectively. Bending test was carried out between 0 and 10 kN loads with 0.5 mm/min displacement rate at room temperature with a dry air.

3. Results

Glass samples showed no color change after the ion exchange process. For all treated samples, UV–Vis spectroscopic measurements revealed a slight decrease in optical transmission from 92% to minimum 90% in visible wavelength.

SEM analysis realized on the surface and cross section of the unprocessed and processed samples showed no morphological alterations, the microstructure still being amorphous as the unprocessed sample. As it is presented in Fig. 1, the weight of the samples increased gradually both by increasing time and increasing temperature.

EDS analysis results of the surface potassium and sodium ion concentrations were presented in Fig. 2. Results reveal that, the potassium concentration initially presents an increasing trend compared to the unprocessed sample and reaches a stable region after a certain duration, as the surface was saturated. This duration shifted to longer times with

increasing temperature. Surface potassium concentration increases with increasing temperature.

Vickers micro hardness results of all samples are presented in Fig. 3 and the data are given in Table 1 for selected samples. Hardness of ion exchanged samples was higher than unprocessed glass for all process conditions. Decreasing hardness was observed with increasing temperature, where lowest hardness was determined at 500 °C and the highest at 425 °C. For lower temperatures of 425 °C and 450 °C, minor hardness variations were observed with increasing time, including a slight decrease and stabilization. For higher temperatures of 475 °C and 500 °C, initially a similar or higher hardness value was observed as with the lower temperatures, but it was followed by a rapid decrease and stabilization with prolonged treatment.

Further investigations were realized on samples processed for 16 h to understand the effect of temperature and processed at 425 °C to evaluate the effect of time.

Results of line scan EDS analysis is given in Table 1 for selected samples and presented in Fig. 4 for samples 425-1, 425-4 and 425-16. Potassium concentration decreased from surface to bulk in all samples and the diffusion depth increased with increasing time. Shorter treated samples 425-1 and 425-4 showed a rapid decrease of potassium concentration and with increasing time the profile presented a smoother distribution of potassium ions, as observed with sample 425-16. Diffusion depth increased with increasing temperature, profiles of samples resembling sample 425-16.

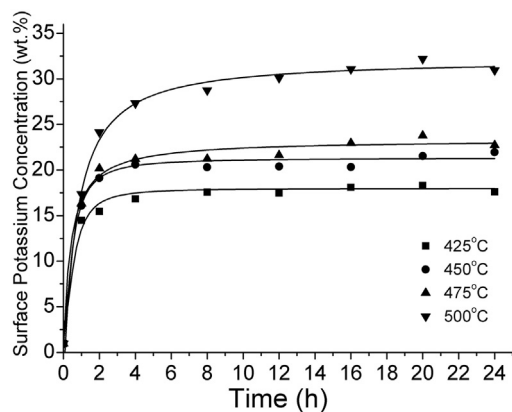


Fig. 2. Surface potassium concentration variations after ion exchange process.

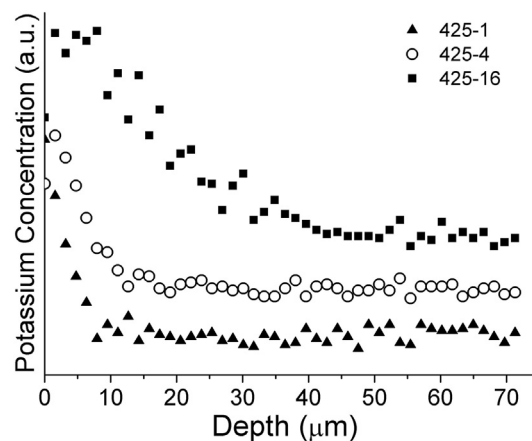


Fig. 4. Line scan EDS results of samples 425-1, 425-4 and 425-16 showing potassium concentration variance with depth.

Table 1

Diffusion depth, Vickers hardness, threshold load for indentation crack initiation (TLICI), crack formation probability, mean crack size, dominant crack type, critical load for scratch cracking and bending strength values of the unprocessed sample and selected samples.

Sample	Diffusion depth (~μm)	Vickers hardness (GPa)	TLICI (N)	Crack formation probability under 9.8 N (1 kg) (%)	Mean crack size under 19.61 N (2 kg) (~μm)	Dominant crack type under 19.61 N (2 kg) (primary + secondary)	Critical load for scratch cracking (~N)	Bending strength (MPa)
Unproc.	–	5.20 ± 0.05	0.98	100	273.07 ± 5.84	Radial + median	3.3	95.77 ± 24.47
425-1	10	5.87 ± 0.05	2.94	90	225.92 ± 6.13	Median + radial	8.2	130.66 ± 20.42
425-4	16	6.08 ± 0.07	9.80	63	202.77 ± 5.43	Median	12.6	334.05 ± 28.26
425-16	42	6.03 ± 0.07	9.80	51	180.51 ± 10.71	Median & lateral	13.0	351.04 ± 32.10
450-16	50	5.91 ± 0.06	9.80	63	180.58 ± 13.13	Lateral + median	13.4	272.34 ± 13.68
475-16	55	5.72 ± 0.05	2.94	100	173.25 ± 15.60	Lateral + radial	10.2	219.98 ± 24.48
500-16	60	5.61 ± 0.08	1.96	100	171.45 ± 9.88	Radial + lateral	7.3	217.26 ± 35.98

Indent and crack images of the unprocessed sample and selected samples obtained under varying loads is given in Fig. 5. TLICI, cracking probability and deformation zone comparisons were realized through the evaluation of these images, combined with the data presented in Table 1.

Deducing from the designated TLICI given in Table 1, the ion exchange process has improved the crack resistance 2–10 times compared to the unprocessed sample depending on the process parameters. The lowest TLICI was observed in highest temperature sample 500-16. Sample 475-16 with slightly higher temperature and sample 425-1 that was shortly ion exchanged presented the second lowest TLICI. Samples 425-4, 425-16 and 450-16 presented identical TLICI. Therefore, for the sake of righteous evaluations on the strength alterations among these samples, further comparisons as cracking probability and deformation zone analysis were determinative.

All samples formed full cracks under 9.8 N and 19.61 N loads. Crack size comparisons were realized under 19.61 N regarding the crack types

[8,13,19] samples formed. Fig. 5 shows that unprocessed sample formed both radial and median cracks dominantly; samples 425-1 and 425-4 formed median cracks with primary dominance whilst 425-16 sample formed both lateral and median cracks with similar dominance. With increasing temperature, lateral cracking gradually became more dominant along with expanding radial surface cracks.

Measured sizes of the cracks under 19.61 N is given in Table 1. Time dependence of crack size was evaluated through samples 425-1, 425-4, 425-16, where crack sizes were determined as the average of median crack lengths. Median crack sizes decreased in the order 1, 4, and 16 h of process time.

The temperature dependence of crack size were evaluated through samples 425-16, 450-16, 475-16 and 500-16, where damaged areas were determined as the average of lateral crack diameters.

Lateral crack sizes decreased with increasing process temperature as the surface radial cracks became visibly larger. From Table 1, it can be commented that ion exchanged samples formed smaller cracks than the

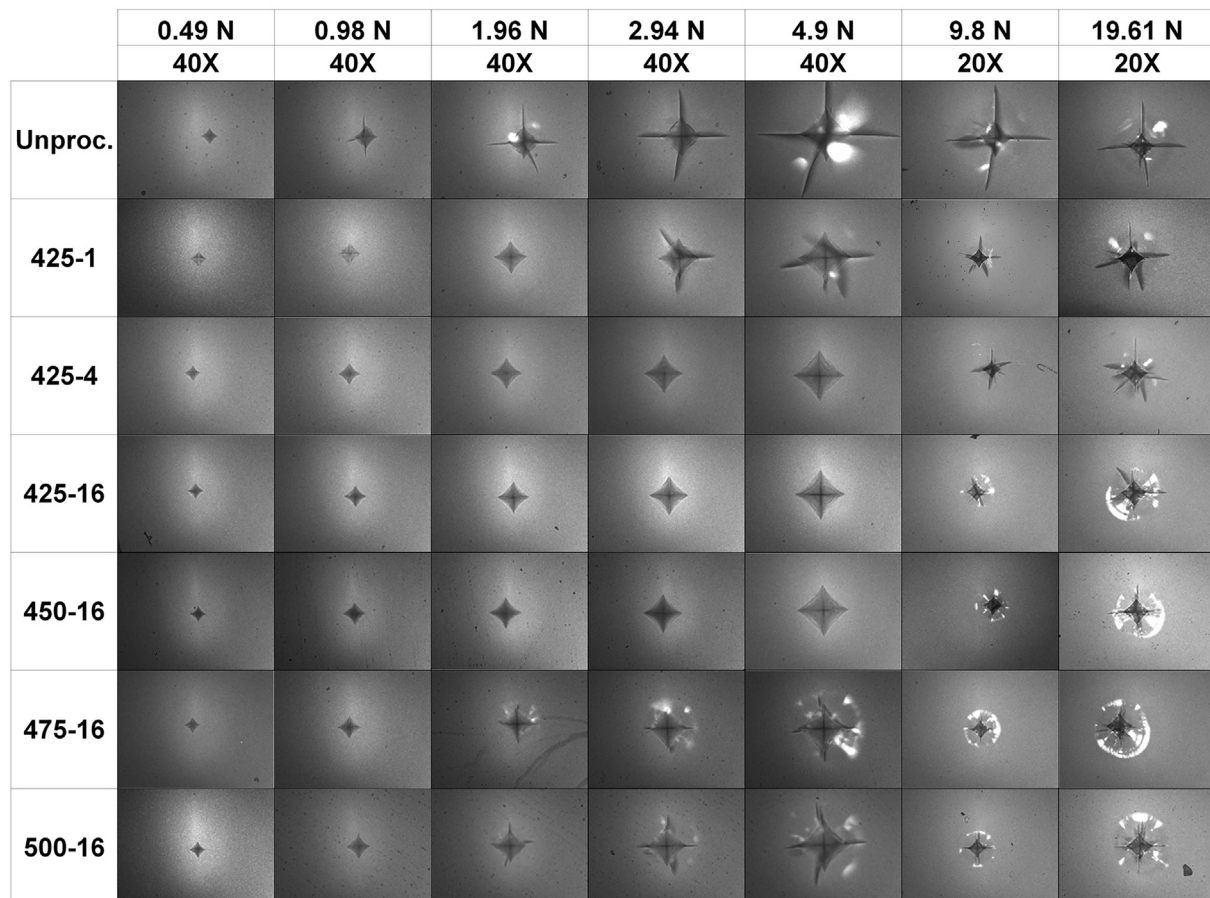


Fig. 5. Indentation and crack images of the unprocessed sample and selected samples obtained under varying loads.

unprocessed sample.

Fig. 6 presents the crack formation probabilities of the samples and data are given in Table 1. After the TLICI limit, which until the cracking probability is zero, all samples presented an increasing probability with increasing load. Unprocessed sample presented the highest probability under each loading condition, compared to the ion exchanged samples. Under constant loading and time, increasing process temperature increased the crack formation probability. For constant temperature, probability was seen to have risen in the order 16, 4 and 1 h of treatment. In addition, samples 425-4 and 450-16 presented very close probabilities which was slightly higher than 425-16 (Fig. 6).

Samples 425-16 and 450-16 with the two lowest cracking probabilities, similarly showing the highest TLICI and lateral cracking, also formed some amount of chipping after indentation under 19.61 N as seen in Fig. 7 with a representative image.

Also it is important to make it clear that only radial cracks are counted for determining crack resistance, because the cracks normal to glass surface are critical to fracture of glass [7,9] so comparison

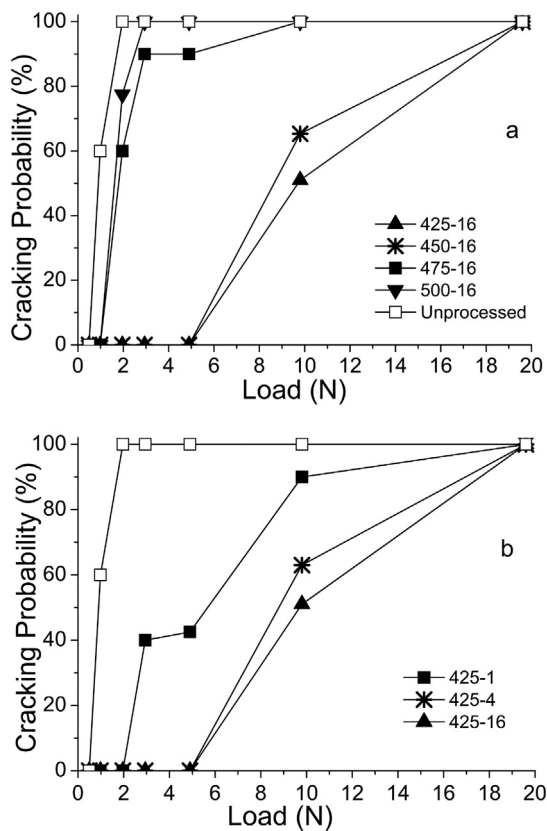


Fig. 6. Cracking probabilities of unprocessed sample and selected samples for a) varying temperatures and b) varying times.

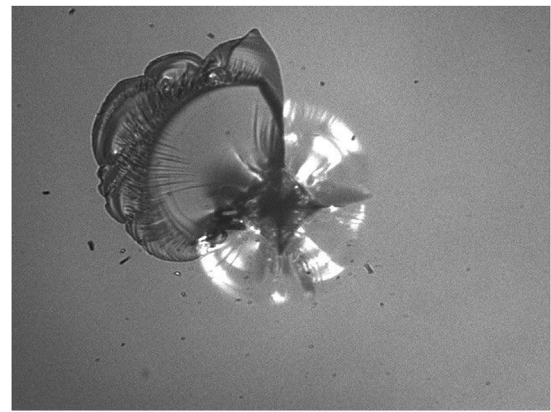


Fig. 7. Chipping formation of sample 425-16 under 19.61 N.

through fracture toughness measurements were not viable.

Scratch tests were carried out to investigate the resistance of ion exchanged glasses to sliding contact. The graph in Fig. 8 displays the scratch test results of the unprocessed sample and selected samples, containing the changes in acoustic emission data against an increasing load and indenter displacement. Critical loads for scratch cracking were determined as the load corresponding to the first observable oscillations in the acoustic emissions [8,17,18,20] and the data are given in Table 1. Observation of the scratch is also a powerful tool for the understanding of crack initiation and amount of damage when evaluated together with acoustic emission signals [18,21].

Until the critical loads are reached, acoustic emission signal is very close to zero as can be seen as a straight line for all samples in Fig. 8 and affirming only a straight residual plastic groove line was observed without the crack formation, as reported for soda lime glass [18,20,21]. This line can be vaguely seen in the first image of Fig. 9a presenting the selected images along scratch for the unprocessed sample under increasing loading from left to right. After that limit, vague Hertzian type

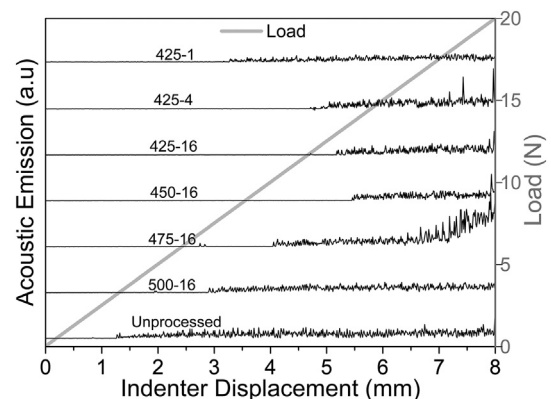


Fig. 8. Scratch test results of the unprocessed sample and selected samples.

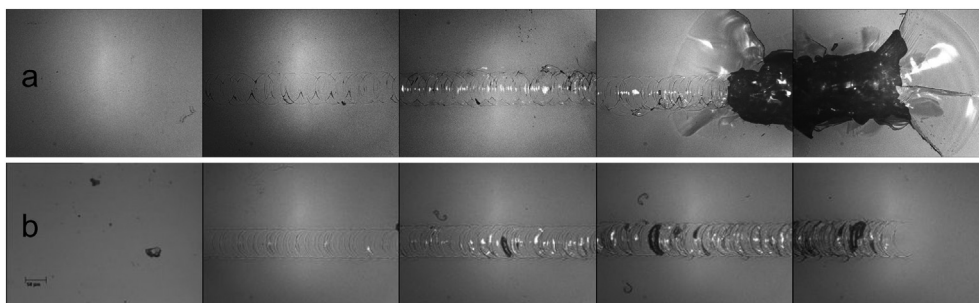


Fig. 9. Scratch images under increasing load from left to right for the a) unprocessed sample and b) processed samples.

ring cracks [17] started to appear and became more distinguishable with increasing load as observed in the second and third images of Fig. 9a. The last two images of Fig. 9a presents the chipping formation observed for unprocessed sample under elevated loads [8]. Chipping occurs at elevated loads when the lateral cracks eventually intersect the surface and remove material [18,21]. The critical scratching load increased after the ion exchange treatment for all selected samples and chipping formation was not observed. All treated samples formed Hertzian type of ring cracks along the scratch and exemplary images of a sample were presented in Fig. 9b. From the acoustic emission signal initiations in Fig. 8, it can be seen that rising process temperature decreased critical load. Among samples treated at 425 °C, 1 h sample showed the lowest critical load, where 4 and 16 h samples presented close and higher values. Overall evaluation of all the selected samples indicate a close critical load for samples 425-4, 425-16 and 450-16.

Strength values obtained by four point bending tests are listed in Table 1. Results indicated that the strength of the samples were increased after ion exchange process. Fig. 10a presents that among samples ion exchanged for 16 h, the strength was deceleratingly decreased by the increase in process temperature. In Fig. 10b, it can be observed that increasing time has led to the decelerating elevation of strength. 425-16 sample had the highest strength as about 3.7 times of the unprocessed sample.

4. Discussion

Improvement of surface and bulk mechanical properties in glasses is directly dependent on the amount of residual compressive stress generated on the surface and the stress profile obtained along the diffusion depth. And this character of compression stress is directly related to the amount of the ions imparted to the surface, ionic diffusion depth and profile. The depth of this compressive stress is called the case depth and stress profile resembles the concentration profile of the penetrating ions

unless relaxation is present. In this regard, adjustment of two opposing processes occurring during ion exchange; stress build up and viscous stress relaxation gains importance to achieve a desired profile [1–3,6,11,16,22–28]. Considering the chemical tempering process, higher potassium (or another larger ion) containing glass surfaces are expected to host higher compressive stress and present better mechanical properties. But if glass suffers any strain limiting structural or viscoelastic relaxation that becomes even more effective with increasing temperature, either bulk relaxation occurs or the localization of compression maximum is shifted from the very surface to microns below due to surface relaxation (subsurface maxima). This shift also can be observed with relatively prolonged treatment due to structural relaxation and/or viscoelastic stress relaxation proceeding in a much slower rate at low temperatures [1–3,11,16,23,28–30]. The result is as mentioned, concentration profile and compressive stress profile may not coincide. Tyagi and Varshneya [16] present various approaches to the causes of the relaxation phenomena. It was believed that it would be possible to comment on the compressive stress levels [14] and relaxation phenomena through evaluating the mechanical test results obtained under present experimental conditions together with the concentration profiles [7,11,13,28,31].

As also observed in studies of Jiang et al. [32], the potassium concentration was stabilized in time, so the surface was gradually saturated. And the increase in the potassium saturation concentration of the surface with increasing temperature was related to further relaxing of surface structure to become more capable of accommodating higher amount of large potassium ions [20]. As the diffusion is a heat driven process proceeding further with time, and structural relaxations are known to occur through glass network at especially elevated temperatures [22], a higher diffusion rate was expected of the ionic exchange process with increasing temperature and time. The measurements revealed that the weight of the samples increased with both increasing parameters, as also mentioned by Gy [1]. So potassium ion incorporation to the sample must have incremented by temperature and proceeded with time, leading to deeper diffusion into the bulk of the glass as affirmed by line scan EDS analysis and also reported by Jiang et al. [32].

The line scan EDS results resemble former studies [15,27] regarding the behavior of concentration profiles with varying time and temperature. The rapidly decreasing concentration profile of 4 h sample was slightly deeper than the 1 h sample, yet surface and bulk mechanical responses were detected much better which was in accordance with the comments of Houérou et al. [20]. Prolonged treatment (in this case 16 h) at low temperatures brought even better overall mechanical properties. Regarding the effect of temperature, higher surface potassium concentration or deeper profile obtained at higher temperatures would not necessarily imply a better mechanical performance at all times as will be discovered shortly [1,28]. That is caused by stress relaxation overcoming the stress build up process at temperatures close to the glass transition temperature, so the very high rate of diffusion remains insignificant [1,25,26].

Houérou et al. [20] explains that the more free space in the glass network is occupied the higher becomes the density, which leads to an increase in hardness. In the same manner, incorporation of larger potassium ions to the formerly sodium ion containing sites [12,31] gradually cause structural strain and thus increase compressive stress, which eventually enhances hardness as the resistance of the surface against indentation is increased [6,7,10,12–14,31,33,34]. Results revealed that the hardness of ion exchanged samples were increased by about 8–17% compared to the unprocessed glass, close to the results of former studies on ALS glass [13] and soda lime glass [10,31]. Viscoelastic relaxation phenomena was believed to be the reason for decreasing hardness with increasing temperature despite the increase in surface potassium concentration and diffusion depth. The dependence of hardness on compressive stress for the present study resemble the findings of Koike et al. [12].

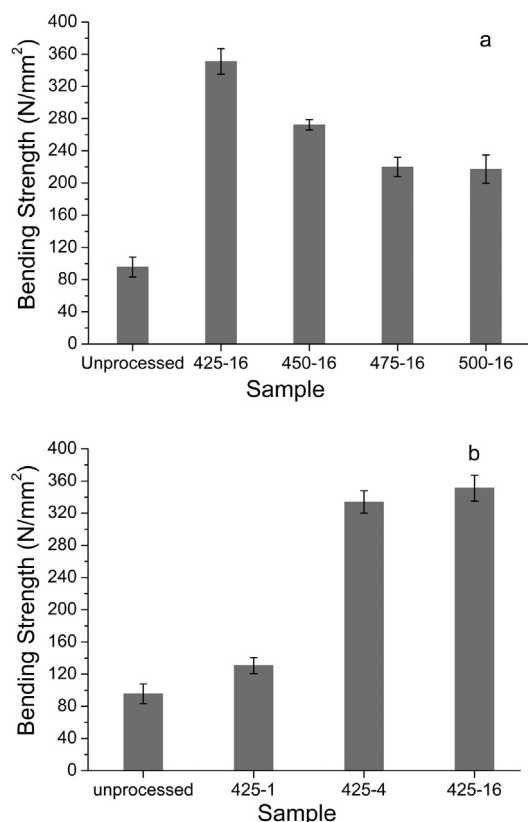


Fig. 10. Strengths of unprocessed sample and selected samples for a) increasing temperature and b) increasing time.

For lower temperatures hardness decrease in prolonged time was insignificant, so was the surface stress relaxation as already known for well below (about 90 K) the glass transition temperature [35–38]. The reason for this slight decrease was believed to be the glass network relaxation due to restructuring or volumetric alteration, both caused by excessive incorporation of the potassium ions to the surface with increasing time [1,14,23,39] and leading to slightly lower levels of compressive stress build up [2,3,12]. Supportingly, in short term, higher temperature processes were observed to result in a higher hardness but the values decreased with time much more rapidly compared to lower temperatures. So increasing ion exchange temperatures cause the rapid decrease of surface stresses [37], thus the decreasing hardness was attributed to faster viscoelastic relaxation and also faster restructuring of the network with the aid of heat. For temperatures towards well below the glass transition, the time necessary to achieve the maximum strength got longer, but the maximum achievable strength got higher [11,29,40].

The function of compressive stress on glass surface is to overcome the residual tensile stresses induced by post Vickers indentation and suppress the crack growth and propagation [12,24,41,42]. Such glasses are also expected to generate smaller indents, cracks and deformation zones than the ones with lower compressive stress, under the same loading conditions [7,12,14,15,31]. From the obtained results of TLICI analysis for all selected samples, it was commented that the generated compressive stress on the glass surface shifted the crack initiation to higher loads improving the mechanical behavior, as was mentioned in former studies [6,12,13,24,25]. Similarly observed by Toth et al. [14] for unprocessed sample, full cracks started to appear around 1 N (100 g) while the processed samples started to develop cracks at minimum 2 N (200 g). After the ion exchange treatment samples presented varying combinations of radial, median and lateral cracks depending on the process parameters, detailed evaluations and comparisons were made in further reading. Some studies [9,11,13,14,21,24,43] also inform about similar crack type variations depending on time, temperature, applied load, stress distribution around the indentation and surface compressive stress level, depth and profile. A higher surface compressive stress and a deeper layer of compression can change the crack type, as the surface response varies thanks to the effective coverage of deeper surface flaws.

The effect of process time on cracking behavior was evaluated by the comparison of TLICI, median crack sizes that samples in question commonly formed and their cracking probabilities. The low TLICI and high cracking probability of shortly treated (1 h) sample clearly indicated an insufficient chemical strengthening compared to longer treated samples (4 and 16 h) that presented similar and higher limits. This similarity was probably be due to the loading range limitation of the instrument used, yet the evaluations made together with other surface and bulk mechanical properties of samples were enough to handle proper comparisons. Gross [21] mentions that the glasses lacking mechanical strengthening form highly radial/median cracks prior to lateral cracks, as did the unprocessed sample and shortly ion exchanged samples. And in highly ion-exchanged samples (in this case 425-16) lateral cracks started to form preferentially, since a surface compression stress with a deeper profile only enable the cracks initiated at the tip of the indenter to expand through a similarly stressed horizontal zone (prevent propagation perpendicular to surface) that does not suppress their extension [15]. It was observed that median crack sizes decreased with increasing time, but at this point it is crucial to note the crack type change of 16 h sample. The occurrence of lateral cracking consumed some of the residual energy that median cracks consumed to grow, thus presenting a lower median crack size when coexisted with lateral cracks [15]. Under constant loading, cracking probability was decreased with increasing time, which was expected as the samples with higher TLICI and generating smaller cracks would be less prone to cracking, and the evaluations were made through the concept of brittleness. Prolonged treatment results in a slight amount of

surface stress relaxation which was observed as a slight decrease in hardness, but at the same time the diffusing ions reach much deeper parts of the surface than the flaw depth, changing the crack type to lateral [1,11,12,24]. Accordingly, larger case depth provides improved cracking behavior [44].

Former studies and the present study revealed that there existed a correlation between hardness and indentation induced cracking properties, which eventually effected the cracking probability. Varshneya [3] defines brittleness as “hardness/toughness” and Koike et al. [12] explains that when K replaces Na, network bonding gets stronger due to the increase in density, which is promoted by the lack of network rearrangement. Moreover, the chemically strengthened glasses act in a brittle way, unless compressive stresses are high enough to obstruct the formation of median cracks. So it can be commented that the level of ionic exchange was enough to increase the hardness but insufficient to impart enough cracking resistance until prolonged treatments. It could also be related to the crack type change, as radial/median type of cracking would obviously promote fracture more than lateral cracking. So high enough hardness with dominantly median type of cracking (sample 425-4) brought along brittleness leading to an increase in cracking probability. And slightly lower hardness with lateral crack dominance (sample 425-16) has decreased the probability of fracture.

On the other hand, over 450 °C increasing temperature decreased the threshold load as did the hardness. Under identical loading conditions, it brought along enlarging radial surface cracks besides contracting lateral cracks and increased cracking probability. This clearly signified the deterioration of surface mechanical properties, since radial cracking preferably proceeds in the absence or deficiency of compression [7,43]. So as suggested in other studies [1,11,29,30,37], it could be commented that the viscoelastic relaxation took place on the surface rapidly at high temperatures, and probably the compression maxima is found below the surface [16].

At highest load, chipping was observed in highly ion exchanged samples (425-16 and 450-16), which were deduced to be so according to other mechanical test parameters (revisit Fig. 7). It was commented that as the high compressive stress level on the surface did not allow the growth of radial or median cracking, residual stresses induced at high loads were efficient enough for the lateral cracks to enlarge and finally intersect the surface to cause chipping [15,21].

It was observed that after ion exchange process strength has led up to around 350 MPa, which was about 3.5–4 times compared to the unprocessed sample, as also was mentioned in literature [11,27,29]. Surface residual compressive stress and depth in ion exchanged glass are directly related to the final strength and fracture pattern of the glass, as the strength of the glass is directly affected from the amount and distribution of flaws on its surface. The glass is known to present a much lower strength than the theoretical value, since the stress concentration at the tip of the cracks (flaws) initiated on the surface cause a fast propagation and finally fracture [9]. Surface flaws are known to possibly extend up to around 30 µm depth [11] so the case depth is supposed to be larger to fully cover the defects and obtain effective strengthening, which requires even deeper diffusion depths. Increasing strength was observed with increasing process time, where simultaneously the median crack size decreased and the dominant crack type shifted from radial/median to lateral thanks to the deepening concentration profile. But this is surely only in the presence of a minor stress relaxation that would cause a deceleration of strength increase at prolonged treatments [44]. The correlations between the crack type and strength for this study are once more supported by Schneider et al. [8] and Gross [21]. Regarding the resemblance of hardness and TLICI test results of samples 425-4 and 425-16, one might have expected similar bending strengths, but as was mentioned by Varshneya [3], strength increase was larger than the hardness increase. And if the line scan EDS analysis (Fig. 3) is revisited, it can once more be seen that sample 425-16 is favored considering the crack type and potassium ion concentration profile.

The reason for decreasing strength with rising temperature was believed to be the increasingly promoted initiation and propagation of radial surface cracks on the surface due to the elevation in stress relaxation at temperatures approaching the glass transition temperature [22–24]. This could be predicted through the TLICI variations depending on process temperature, so the higher the temperature was, the lower would be the maximum strength to be achieved [11,29,45].

Fracture evaluations of bending tests showed that shorter treated samples with tendency of radial/median cracking and shallower diffusions depths (thus case depths) presented an easier fracture and less fragmentation. So as can be predicted, unprocessed sample presented a sudden fracture and least fragmented visual. With increasing lateral cracking tendency maintained via longer treatments, a smaller fragmentation and a more flexible behavior until fracture was observed. This behavior was similar to that of thermally tempered glasses possessing high interior tensile stresses, which can be achieved with chemical tempering at prolonged treatments [46]. Eventually, the strength variations related to the concentration profiles were in accordance with the crack formation probabilities and varying crack types.

Scratch resistance of the treated glasses were compared regarding their critical loads that initiated damage on sample surface. As similarly observed for other surface mechanical properties, scratch resistance was generally enhanced with increasing time, as did the cracking properties. Low temperature treatments resulted in close scratch resistances, while above 450 °C with increasing temperature, gradual deterioration of scratch resistance was detected. This was attributed to stress relaxation of the surface at temperatures approaching to glass transition, which eventually promoted the formation of radial type of surface cracks. Gross [21] supports that as the TLICI of median/radial cracking increases, so does the critical load for scratch lateral cracking since the cracking mechanisms under the subsurface highly resemble each other. It is apparent that the less tendency a sample would have for cracking, the harder it would be for ring cracking to occur on the surface during scratch. The same study claims that scratch visual lateral cracking threshold is not related to the compressive stress profile in case of highly ion exchanged samples where the lateral type cracking is favored instead of median/radial cracking. The close thresholds of lowest temperature treatments may be related to this claim, since relaxation was not considered as an effective mechanism at those temperatures.

5. Conclusion

There was no significant optical transmittance difference of the samples after the ion exchange process. Surface EDS analysis showed that the surface was saturated with ions after a short while and the saturation concentration increased with increasing temperature. Line scan EDS analysis revealed a stable decrease of potassium concentration from surface to bulk for all samples for all process parameters. Increasing diffusion depth was observed for increasing time and temperature, since the weight also increased despite surface saturations. Compared to unprocessed sample, ionic exchange improved the micro hardness by 8–17%, TLICI by 2–10 times, cracking probability down to 50%, strength up to 3.5–4 times and scratch resistance 2–4 times. Increasing temperature deteriorated the mechanical properties, while the true effect of time on mechanical improvement had to be evaluated from a variety of perspectives. With time, hardness initially increased up to a maximum value and later descended to become stabilized. Towards lower temperatures, maximum hardness value shifted to longer treatments and the descent with increasing time could be considered very small. At higher temperatures, hardness decreased rapidly due to faster viscoelastic relaxation. Increasing hardness was observed towards lower treatment temperatures. Ion exchange improved the TLICI for all treatment parameters. Increasing temperature decreased the threshold load. Righteous evaluation of the time effect on surface mechanical response for similar TLICI samples was possible through the integrated investigation of hardness, cracking probability and crack

type variations. If ionic exchange levels were enough to effectively increase surface compression and thus hardness, and yet the diffusion profiles were still to promote radial/median cracking, high tendency of brittleness brought along high cracking probabilities. For slightly lower hardness obtained through prolonged treatment, a deeper diffusion promoted lateral cracking and the probability was decreased with decreasing brittleness.

For all temperatures, high amount of ion exchange through prolonged treatment changed the indentation induced crack type from median/radial to lateral, thanks to a deeper diffusion. At temperatures close enough to glass transition, relaxation was effective enough to lower the threshold load. Until the occurrence of considerable relaxation with temperature, cracking probability and hardness were deteriorated only slightly for identical threshold load.

Scratch resistance was improved after ion exchange. Similar critical load for scratch cracking was observed for similar TLICI, and scratches contained Hertzian type ring cracks. Prolonged treatment increased the scratch resistance since indentation induced crack type changed from median to lateral. Due to the relaxation, increasing temperature deteriorated scratch resistance.

Prolonged ion exchange enhanced strength up to 3.5–4 times at low process temperatures, where strength increased with increasing time. At higher temperatures, strength decreased as radial cracking was gradually promoted with elevating stress relaxations.

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